3.3.2 Verlet

Recall the position and velocity Verlet methods:

$$\begin{aligned} &\textbf{Position Verlet}: && \textbf{Velocity Verlet}: \\ &\hat{\boldsymbol{Q}} := \boldsymbol{q} + (h/2)\, \boldsymbol{M}^{-1}\boldsymbol{p}, && \hat{\boldsymbol{P}} := \boldsymbol{p} - (h/2)\, \nabla U(\boldsymbol{q}), \\ &\boldsymbol{P} := \boldsymbol{p} - h\, \nabla U(\hat{\boldsymbol{Q}}), && \boldsymbol{Q} := \boldsymbol{q} + h\boldsymbol{M}^{-1}\hat{\boldsymbol{P}}, \\ &\boldsymbol{Q} := \hat{\boldsymbol{Q}} + (h/2)\, \boldsymbol{M}^{-1}\boldsymbol{P}. && \boldsymbol{P} := \hat{\boldsymbol{P}} - (h/2)\, \nabla U(\boldsymbol{Q}). \end{aligned} \tag{3.3}$$

We can think of the velocity Verlet method as being defined by a splitting into three parts:

$$\begin{aligned} H(\boldsymbol{q},\boldsymbol{p}) &= H_1 + H_2 + H_3, \\ H_1 &= \frac{1}{2}U(\boldsymbol{q}), \\ H_2 &= \frac{1}{2}\boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p}, \\ H_3 &= \frac{1}{2}U(\boldsymbol{q}). \end{aligned}$$

This is a sequence of two types of operations: it consists of a "kick" exhibited by a jump in the momentum, linear "drift" with the resulting momentum, followed by a final kick. The symmetry of the method is one of its important features. Switching the order of the operations, i.e. drift-kick-drift, gives the position Verlet method.

In general for the Verlet method applied to Hamiltonians of the form $H(\boldsymbol{q},\boldsymbol{p}) = T(\boldsymbol{p}) + U(\boldsymbol{q})$, the BCH lemma gives

$$\begin{aligned} \tilde{H}_h = T + U &+ \frac{h^2}{12}\left(\{T,\{T,U\}\} - \frac{1}{2}\{U,\{U,T\}\}\right) \\ &+ \frac{h^4}{120}\left(-\frac{1}{6}\{T,\{T,\{T,\{T,U\}\}\}\} + \frac{1}{3}\{U,\{T,\{T,\{T,U\}\}\}\}\right. \\ &\left. -\frac{1}{4}\{U,\{U,\{T,\{T,U\}\}\}\} + \{T,\{T,\{U,\{U,T\}\}\}\}\right) + \mathcal{O}(h^6) \end{aligned}$$

The extra power of h in the leading perturbation reflects the fact that this method is 2nd order accurate, rather than first order, as Symplectic Euler. Note that only even order terms $(h^2, h^4, \ldots)$ appear in the perturbative expansion; this is a consequence of the symmetry of the method.